Project Plan: Detecting Out-of-Context Image-Caption Misinformation with Multimodal Embeddings

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1. Background

Out-of-context (OOC) misinformation, a real and unaltered photograph paired with a caption that misrepresents what it shows, is a documented and common tactic in modern disinformation campaigns. In a large-scale study of political WhatsApp groups in India, out-of-context images accounted for 34% of all misinformation images shared, the single largest category, ahead of doctored memes (30%) and digitally manipulated photos (10%) [5]. The same pattern shows up outside social media: among the false rumors that circulated after Iran's January 2020 attack on U.S. bases in Iraq, 12 of 22 documented claims involved real photos or videos shown out of context [4]. Because the image itself is never manipulated, standard image-forensics tools find nothing wrong with it, and text-only fact-checking tools have no mechanism to compare a claim against the specific image it accompanies. This gap has motivated a growing body of multimodal machine learning research aimed at automatically detecting when an image and its caption are semantically inconsistent.

The technical foundation for this project is CLIP (Contrastive Language-Image Pretraining), a model trained on 400 million image-caption pairs scraped from the internet to embed matching images and text into a shared vector space [1]. Building on this idea, the NewsCLIPpings dataset was constructed specifically to benchmark out-of-context detection: it takes real news images and captions and automatically generates falsified pairs through several retrieval methods, producing labeled splits such as a person-match split (swapping in a caption about a different named person), a scene-match split (swapping in a visually similar image to misrepresent the event), and semantic-match splits based on CLIP or sentence-embedding similarity between the image and caption [2]. A related benchmark, COSMOS, takes a self-supervised approach to the same underlying problem and provides a useful point of comparison for how well findings generalize beyond NewsCLIPpings' own generation methodology [3].

This project is academically motivated: it applies an established multimodal embedding technique to a well-defined, publicly benchmarked problem, with the goal of producing both a rigorously evaluated model and a usable demonstration of how such a system would work in practice.

2. Project Aims and Objectives

The project aims to build, evaluate, and deploy a system that detects out-of-context image-caption pairs using multimodal embeddings, and to identify which categories of mismatch are hardest for such a system to catch. Specific objectives:

- Build a data pipeline that acquires, validates, and structures a usable subset of the NewsCLIPpings benchmark for local development.
- Develop and compare at least three levels of model sophistication: a similarity-threshold baseline, a logistic regression classifier, and a lightweight neural network, all built on top of pretrained CLIP embeddings.
- Quantitatively evaluate model performance overall and broken down by mismatch category, including statistical comparison across model variants.
- Deploy a live, interactive application that allows a user to submit an image and caption and receive a real-time consistency assessment.
- Communicate the system's capabilities and limitations clearly through documentation, a live demo, and a visual poster.

Why existing tools cannot meet these goals

- Reverse image search tools (e.g., Google Images, TinEye) can show where an image has previously appeared online, but they do not score whether a specific new caption is semantically consistent with the image, and they only work if the image was already indexed somewhere findable.
- Image forensics tools (error-level analysis, manipulation detectors) only catch pixel-level tampering. Since out-of-context misinformation uses a genuine, untouched image, these tools report nothing wrong and can create a false sense of security.
- Text-only fact-checking tools evaluate the plausibility of a written claim in isolation and have no mechanism to cross-reference it against the accompanying image.
- No existing consumer-facing tool combines image and text understanding into a single, real-time consistency score in an accessible interface, which is the specific gap this project addresses.

3. Expected Outcomes Outline

This project aims to answer the following questions:

- Can CLIP-based embedding similarity reliably distinguish genuine image-caption pairs from out-of-context pairs in the NewsCLIPpings benchmark?
- Does a trained classifier (logistic regression, then a lightweight neural network) built on top of CLIP embeddings outperform a simple similarity-threshold baseline, and by how much?
- Which of NewsCLIPpings' mismatch-generation methods (person-match, scene-match, or semantic/CLIP-based match) are hardest for the model to detect, and what does that suggest about the limits of embedding-based detection?
- How well does the model's confidence calibrate with its actual correctness, i.e., are its high-confidence predictions actually more reliable than its low-confidence ones?
- Can the model's decision be made interpretable to a non-technical user, for example by surfacing similar genuine pairs for comparison, in a way that builds appropriate (not blind) trust in the tool?
- How does model performance change with the size of the training subset, and does that inform whether the planned data acquisition scope is sufficient?

4. Potential Datasets and Resources

NewsCLIPpings github.com/g-luo/news_clippings (also mirrored at huggingface.co/g-luo/news-clippings)

Primary benchmark dataset. Provides labeled genuine and falsified image-caption pairs across defined mismatch-generation-method subsets (person-match, scene-match, and semantic-match); will serve as the core training and evaluation dataset.

VisualNews github.com/FuxiaoLiu/VisualNews-Repository

The large-scale news image-caption corpus that NewsCLIPpings draws its underlying images and captions from. Needed to resolve and download the actual source images referenced by NewsCLIPpings' annotation files.

OpenAI CLIP github.com/openai/CLIP (model weights via huggingface.co/openai/clip-vit-base-patch32)

The pretrained multimodal embedding model used to generate joint image and text representations. Used off-the-shelf via transfer learning rather than trained from scratch, which keeps the project's scope realistic.

COSMOS dataset github.com/shivangi-aneja/COSMOS

A related out-of-context misinformation benchmark. Will be used as a supplementary robustness check to see whether findings generalize beyond NewsCLIPpings' specific mismatch-generation methodology.

Papers with Code (NewsCLIPpings leaderboard) paperswithcode.com/dataset/newsclippings

Reference point for comparing this project's baseline and trained model performance against previously published benchmark results on the same dataset.

5. Evaluation Criteria

The project will be considered successful if it meets the following criteria:

- **Model performance:** the trained classifier outperforms the naive similarity-threshold baseline by a meaningful margin and achieves strong discrimination (target ROC-AUC above 0.80) on held-out test data.
- **Interpretable failure analysis:** the project clearly demonstrates and statistically supports which categories of mismatch are hardest to detect, rather than reporting only a single aggregate accuracy number.
- **Functional deployment:** the live demo application reliably accepts a real image and caption, returns a result within a few seconds, and handles common edge cases (unusual image formats, empty or very short captions) without crashing.
- **Reproducibility:** the data acquisition, preprocessing, and modeling pipeline are documented clearly enough that the process could be repeated or audited from the code repository.
- **Honest scope communication:** the final documentation and presentation transparently state the system's limitations, such as dataset-specific biases or generalization limits beyond the benchmark's mismatch-generation methods, rather than overselling accuracy.

These will be assessed through held-out test set metrics (precision, recall, F1, ROC-AUC), the category-wise breakdown analysis, live testing of the demo application, and review of the final documentation against the weekly milestone tracker described below.

6. Project Plan and Timing

The timeline below follows the same 14-week structure as the course's own project-plan template, with Week 1 beginning September 13, 2026. Dates are planning estimates and will be reconciled against the official course calendar as deadlines are confirmed.

Week	Milestone / Deliverable	Activities
1	Project kickoff & refinement	Review the approved proposal; finalize the project objectives and the CLIP-embedding-based detection methodology.
2	Dataset exploration & acquisition	Identify and begin acquiring data from NewsCLIPpings and VisualNews; review COSMOS as a candidate secondary dataset. Project Plan due this week.
3	Data preprocessing	Clean, validate, and structure the curated image-caption subset; resolve broken image URLs and set up local dataset storage. Design Document due this week.
4	Exploratory embedding analysis	Generate initial CLIP embeddings; analyze similarity-score distributions and cosine similarity across genuine and falsified pairs.
5	Baseline modeling	Build and evaluate a similarity-threshold baseline classifier for genuine vs. out-of-context pairs.
6	Visualization of categories and trends	Create embedding-space scatter plots, similarity-score distributions, and breakdowns by mismatch-generation method (person-match, scene-match, semantic-match).
7	Midterm progress report	Submit a summary of findings so far; receive instructor feedback on baseline results and modeling direction.
8	Classifier development - Part I	Train and tune a logistic regression classifier on the CLIP embeddings; compare performance to the baseline.
9	Classifier development - Part II	Train and tune a lightweight neural network classifier; perform a per-category performance gap analysis against the logistic regression model.
10	Optional: secondary generalization check	If time allows, evaluate the trained classifier on the COSMOS dataset to test generalization beyond NewsCLIPpings' mismatch-generation methodology.

Week	Milestone / Deliverable	Activities
11	Application interface development	Build out the Streamlit demo interface (image upload, preset examples, results panel) and integrate the best-performing trained classifier for live inference.
12	Report writing & visual storytelling	Draft the final report and documentation; refine the evaluation dashboard and embedding visualizations.
13	Slide deck preparation & report revision	Create the final presentation slides and poster; finalize the written documentation and code repository.
14	Final presentation & submission	Deliver the final presentation and submit the Final Project and Documentation.

Weekly Meetings and Progress Reports will run throughout every week listed above and are not a separate phase; each week's report will reference the specific activity completed that week from this table.

References

[1] Radford, A., Kim, J. W., Hallacy, C., et al. (2021). Learning Transferable Visual Models From Natural Language Supervision. Proceedings of ICML 2021. https://arxiv.org/abs/2103.00020

[2] Luo, G., Darrell, T., & Rohrbach, A. (2021). NewsCLIPpings: Automatic Generation of Out-of-Context Multimodal Media. Proceedings of EMNLP 2021. https://arxiv.org/abs/2104.05893

[3] Aneja, S., Bregler, C., & Niessner, M. (2021). COSMOS: Catching Out-of-Context Misinformation with Self-Supervised Learning. arXiv:2101.06278. https://arxiv.org/abs/2101.06278

[4] Fazio, L. (2020, February). Out-of-context photos are a powerful low-tech form of misinformation. The Conversation. https://theconversation.com/out-of-context-photos-are-a-powerful-low-tech-form-of-misinformation-129959

[5] Garimella, K., & Eckles, D. (2020). Images and misinformation in political groups: Evidence from WhatsApp in India. Harvard Kennedy School Misinformation Review. https://doi.org/10.37016/mr-2020-030